main.r

Run ▶

```r
# Online R compiler (editor) for free
# Write and run R online using this editor.
A <- c(18, 2, 20)
B <- c(22, 28, 10)
C <- c(20, 40, 40)

cat("Covariance between B and C:\n")
print(cov(B, C))

cat("Covariance matrix:\n")
print(cov(data.frame(A, B, C)))

cat("Correlation between B and C:\n")
print(cor(B, C))

cat("Correlation matrix:\n")
print(cor(data.frame(A, B, C)))

cat("Chi-square test for age and preference:\n")
tab <- rbind(
  `5-6` = c(18, 22, 20),
  `7-8` = c(2, 28, 40),
  `9-10` = c(20, 10, 40)
)
colnames(tab) <- c("A", "B", "C")
print(chisq.test(tab, correct = FALSE))
```

Output

```
Covariance between B and C:
[1] -20
Covariance matrix:
          A   B         C
A  97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
Correlation between B and C:
[1] -0.1889822
Correlation matrix:
           A          B          C
A  1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
Chi-square test for age and preference:

    Pearson's Chi-squared test

data:  tab
X-squared = 29.603, df = 4, p-value = 5.895e-06
```

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```r
print(bins)

# Smoothing by bin means
smooth_mean <- lapply(bins, function(b) {
  rep(mean(b), length(b))
})

cat("Smoothed using bin means:\n")
print(smooth_mean)

# Smoothing by bin boundaries
smooth_boundary <- lapply(bins, function(b) {
  lo <- min(b)
  hi <- max(b)
  sapply(b, function(v) {
    if (abs(v - lo) <= abs(v - hi)) lo else hi
  })
})

cat("Smoothed using bin boundaries:\n")
print(smooth_boundary)

# Histogram
hist(x, breaks = 10,
     main = "Histogram of Item Prices",
     xlab = "Price",
```

Output

```
Equal-frequency bins:
$`1`
 [1]  1  1  5  5  5  5  5  8  8  8 10 10 10 10 12 14 14

$`2`
 [1] 14 15 15 15 15 15 15 18 18 18 18 18 18 20 20 20

$`3`
 [1] 20 20 20 20 21 21 21 21 25 25 25 25 25 28 28 30

Smoothed using bin means:
$`1`
 [1] 7.705882 7.705882 7.705882 7.705882 7.705882 7.705882
7.705882 7.705882
 [9] 7.705882 7.705882 7.705882 7.705882 7.705882 7.705882
7.705882 7.705882
[17] 7.705882

$`2`
 [1] 17 17 17 17 17 17 17 17 17 17 17 17 17 17 17 17

$`3`
 [1] 23.4375 23.4375 23.4375 23.4375 23.4375 23.4375 23.4375
23.4375 23.4375
[10] 23.4375 23.4375 23.4375 23.4375 23.4375 23.4375 23.4375
```

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Run ▶

```r
A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
B <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)

cat("Class A\n")
cat("Mean:", mean(A), "\n")
cat("Median:", median(A), "\n")
cat("Range:", diff(range(A)), "\n")

cat("\nClass B\n")
cat("Mean:", mean(B), "\n")
cat("Median:", median(B), "\n")
cat("Range:", diff(range(B)), "\n")

boxplot(A, B,
        names = c("Class A", "Class B"),
        main = "Year 9 Exam Results",
        ylab = "Marks",
        col = c("lightblue", "lightgreen"))
```

Output

```
Class A
Mean: 65.77778
Median: 66
Range: 60

Class B
Mean: 62.11111
Median: 60
Range: 34
ERROR!
Error in (function (file = if (onefile) "Rplots.pdf" else
"Rplot%03d.pdf",  :
  cannot open file 'Rplots.pdf'
Calls: boxplot -> boxplot.default -> do.call -> par -> <Anonymous>
Execution halted

=== Code Exited With Errors ===
```

main.r

Run ▶

```r
x <- c(200, 300, 400, 600, 1000)

# (a) Min-max normalization from 0 to 1
minmax <- (x - min(x)) / (max(x) - min(x))

cat("Min-max normalization:\n")
print(minmax)

# (b) Z-score normalization
zscore <- (x - mean(x)) / sd(x)

cat("Z-score normalization:\n")
print(round(zscore, 4))

# Example from the question:
v <- 80000
minimum <- 50000
maximum <- 100000

normalized_v <- (v - minimum) / (maximum - minimum)
cat("Normalized value of $80000:", normalized_v, "\n")
```

Output

```
Min-max normalization:
[1] 0.000 0.125 0.250 0.500 1.000
Z-score normalization:
[1] -0.9487 -0.6325 -0.3162  0.3162  1.5811
Normalized value of $80000: 0.6
```